Business Canvas Project — Assignment 2.3

Unit 2 Tasks 3–4: Primary Research on Customer Preferences + Secondary Research on the Competitive Landscape + Appendix Data Visualization · AP Topic 2.3

Due: _____
Team size: Same team as Assignments 1A, 1.3, 1.4-1.5, 1C, 2.1, and 2.2 (up to 3)
Submission: Three things, all from real data you collected yourself — **(1)** a Business Canvas **Customer** section with **research findings on customer preferences** from your own primary research (Task 3, asterisked); **(2)** a **Market** section with **research findings on the competitive landscape** from secondary sources (Task 4); **(3)** one **Appendix data visualization** (bar / stacked bar / line / pie) that reflects an actual finding from (1) or (2), properly labeled and cited

This is the assignment where the project stops being a pitch and starts being **evidence.** Everything you have written so far — the problem, the profile, the tactic — was a claim. Here you collect data that could have proven you wrong, and report what it said.

The cautionary case is New Coke (1985). Coca-Cola spent **$12 million** (today's dollars) testing with **200,000 people.** New Coke won **every** taste test. They launched it. **Three months later they pulled it**, after 8,000 angry calls a day and demonstrations — and the president said on the record: "*We did not know what we were selling.*"

They were not lazy. They were not stupid. They asked the **wrong question at enormous scale.** Every section below exists to keep you from doing the same thing at small scale.

Review first: [[2.3 Market Research - Vocab]] · [[2.3 Market Research - Slides]] · [[2.3 Market Research - Summary]] · [[2.3 New Coke - Case Handout]] · [[2.2 Consumer Behavior - Vocab]]

Deliverables

Canvas Section	What You Submit	AP Topic	CED Task
Customer	**Research findings on customer preferences** — 2 tested hypotheses, the instrument, the raw data, the verdict, and what changed	2.3	Unit 2 Task 3 · LO 2.3.C
Market	**Research findings on the competitive landscape** — ≥4 cited secondary sources, a competitor table, market size/ trends/segments, and the gaps left over	2.3	Unit 2 Task 4 · LO 2.3.B
Appendix	**One data visualization** (bar / stacked bar / line / pie) built from your own research finding, with title, labeled axes + units, source, and n	2.3	Unit 2 Task 4 · LO 2.3.D

> **NOTE — *Customer – Research findings on customer preferences* is asterisked in the CED: it requires hypothesis testing.** Task 3 is the graded core of this assignment. If you can point to **one number you did not expect**, you have done it; if every result matched what you hoped for, the exam reader will assume you asked leading questions — which is exactly what Section 4 makes impossible.

> **We run Task 4 (secondary) first, on purpose.** The CED lists Task 3 before Task 4; the method runs the other way. **Secondary before primary** — it is cheaper, faster, and it tells you what question is actually worth paying to ask.

Where This Fits — Project Progress

#	Checkpoint	Topics	Assignment	Status
1	Customer problem + competitors + competitive advantage plan	1.1, 1.2	Assignment 1A	✅ Done
2	PESTEL analysis + market verdict	1.3	Assignment 1.3	✅ Done
3	Validation interviews + product idea + hypothesis test + vision & mission	1.4, 1.5	Assignment 1.4-1.5	✅ Done
4	Production process + supply chain plan	1.8	Assignment 1C	✅ Done
5	Commit to the idea + revised market analysis + integrated canvas	1.1–1.8	Unit 1 Final Canvas	✅ Done
6	Target customer profile + plan to build customer relationships	2.1	Assignment 2.1	✅ Done
7	Sales tactic drawing on principles of persuasion	2.2	Assignment 2.2	✅ Done
8	**Primary + secondary research, competitive landscape, data visualization**	**2.3**	**This assignment**	🔵 Current
9	MVP, value proposition, branding	2.4	Assignment 2.4	⬜
10	Pricing strategy	2.5		⬜
11	Marketing channel(s)	2.6		⬜
12	Marketing campaign + digital marketing tools	2.7		⬜

> **What just fed in:** Assignment 2.2 Section 3 gave you a drafted tactic, and its Step 5 (optional) gave you an untested "which wording wins" tally. **Those are hypotheses now.** Bring them here.
>
> **What this feeds forward:** 2.4 needs an **MVP that tests product-market fit**, a **value proposition**, and **branding** — and all three are stronger when they rest on findings you wrote down here instead of on opinion. The **Appendix data visualization** you build in Section 6 is a permanent project deliverable: it travels with you to exam day.

Suggested pacing (≈6 class periods, matching the CED's 6 periods for Topic 2.3):

Milestone	By	You bring to class
M1 — Claim inventory	Day 1	Section 1 complete: what we believe, what we can prove, the one claim we most fear testing
M2 — Secondary scan done	Day 2	Section 2: ≥4 cited sources (≥1 government or academic) + competitor landscape table + the **gap list**
M3 — Hypotheses + instrument approved	Day 3	Section 3: 2 hypotheses in strict form with pre-set success criteria, chosen methods justified, and **every leading question rewritten** — peer-approved before any data is collected
M4 — Fieldwork done	Day 5	Section 4: raw response log, results table, verdicts
M5 — Visualization drafted	Day 5	Section 5: one chart, correct type for its question, fully labeled
M6 — Submit	Day 6	Sections 4–6 complete, New Coke audit (Section 5.3) filled, full worksheet + raw data attached

> **Do not collect a single response before M3 is approved.** Data collected on an unexamined instrument is the most expensive kind of nothing. New Coke's $12 million is the proof.

Before You Start: The Framework

Eight rules from the 2.3 lesson that this worksheet runs on:

1. **Market research = systematically gathering, analyzing, and interpreting information about a market — customers *and* competitors — to guide decisions** (EK 2.3.A.1). Its job is to convert "*I think*" into "*the data shows.*" If a piece of research cannot change a decision you were about to make, do not pay for it.
2. **Two data types, and you need both** (EK 2.3.A.2.i–ii). **Quantitative** = numbers you can count and graph ("*72% prefer X*" — answers *how many?*). **Qualitative** = words, images, observations ("*I like X because…*" — answers *why?*). **Quantitative identifies the pattern; qualitative explains it.**
3. **Three gates, in order** (EK 2.3.A.3.i–iii): **Desirability** (do customers actually want it — problem-solution fit) → **Feasibility** (can we build and deliver it with our resources, tech, time) → **Viability** (does it produce revenue greater than cost over time, sustainably). A product can be desirable but not feasible; feasible but not viable. **All three must pass, and desirability is checked first.**
4. **Secondary before primary** (EK 2.3.B.2–B.3). **Secondary = data that already exists** (government, commercial, academic, internal), collected by someone else for a different purpose. It is **cheap, fast, broad** — cheap reconnaissance that reveals PESTEL factors, market size, trends, customer segments, and competitor positioning. Its weaknesses: it may not answer *your* question, it may be out of date, and **you did not control how it was collected.**

5. **A business hypothesis is an assumption stated so it can be tested** (EK 2.3.C.2) — written **before** the data, in **variable → measure → expected number** form. It does three jobs: it **focuses** what to measure, makes results **interpretable** (supported / not supported), and **prevents post-hoc rationalization.** *A hypothesis is not a wish.*
6. **Match the method to the data the hypothesis needs** (EK 2.3.C.3–C.7). Don't use a survey to answer a *why* question. Don't use a focus group to count behavior. **A/B testing is the gold standard for "will they actually buy?"** because the answer comes from behavior, not from opinions about behavior.
7. **Honest data has two pillars** (EK 2.3.C.8): a **sample large enough and representative enough** for the population you actually care about, and **neutral question phrasing.** A biased question with 1,000 respondents gives you 1,000 useless answers; a neutral question with 15 carefully chosen respondents gives you 15 honest ones. **And always report the base** — 30 of 120 is 25%, not 15%.
8. **Research reduces uncertainty; it does not eliminate it** (EK 2.3.C.9). A survey showing 72% would buy does not mean 72% **will** buy. A 40-point gap over two Saturdays does not prove causation. **Each layer adds confidence; none guarantees the future.** Part of your grade in Section 4 is naming what your own test **cannot** claim.

Chart selection is decided by the question, never by the look (EK 2.3.D.1–D.3):

If you want to show…	Use	Watch out
Values compared across categories	**Bar chart**	x = categories, y = the measured value; compare by height — the workhorse of the exam
Total **and** composition within each category	**Stacked bar chart**	bar height = the total; segments = what's inside
Change over time	**Line graph**	**time goes on the x-axis**
Proportions of a whole	**Pie chart**	best with **3–5 slices**; more than 5 becomes unreadable
Relationship between two measured variables	*(scatter plot — not one of the four Appendix types)*	reveals clusters and dead zones; use it in Section 2, not in the Appendix

> **Strong-answer method (all through this worksheet):**
> **State the finding as a number with its base → name the method and EK that produced it → say what decision changed because of it.**

SECTION 1: THE CLAIM INVENTORY — What Do You Actually Know?

AP Topic 2.3 · LO 2.3.A · EK 2.3.A.1, 2.3.A.2.i, 2.3.A.2.ii, 2.3.A.3.i–iii

Before designing any research, list what the project is currently **assuming.** This is the difference between research and theater.

1A. Believed vs. proven

#	A claim your canvas currently rests on	Where it came from	Status: proven / hinted / pure belief	What would prove it wrong
1	*(e.g. "Parents will pay for a documented inspection")*	1.4-1.5 interviews / 2.1 profile / 2.2 tactic / gut		
2				
3				
4				
5				

Model, from the lesson: Theo's claim "*buyers will pay more for a bike with a documented inspection tag*" started as a belief earned from **one $55 sale.** It became testable only when he wrote it as *"**If I add an inspection tag, at least **60%** of bikes will sell within one week at **$95**."*

Circle the one claim above you most fear testing. That is your Task 3 hypothesis. A claim nobody can hurt is not a claim worth researching.

1B. Sort your existing evidence by data type

Your evidence so far	Quantitative or qualitative?	What it answers	What it cannot answer
		how many? / why?	

The New Coke gap, in one row: Coke's 200,000-person study was **quantitative** about **taste**. The thing that actually decided the outcome — emotional attachment to the brand — was **qualitative** and **never asked**. Fill the right-hand column honestly; it is where your research questions come from.

1C. The three gates — your current verdict

Gate (EK 2.3.A.3)	The question	Your evidence right now	Verdict: pass / fail / unknown
Desirability	Do customers want this? (problem-solution fit)		
Feasibility	Can we deliver it — resources, technology, time?		
Viability	Can it make money, sustainably?		

> **Order matters.** Desirability first: a feasible, profitable product nobody wants is a very efficient way to make nothing. **This assignment is a desirability + landscape exercise.** Viability gets its numbers in 2.5 (pricing) and 3.x (finance).
>
> **Research is not a one-time event (EK 2.3.A.4).** Pre-launch you test the three gates; post-launch you **track changes** (buyer tastes drift from mountain bikes to commuters), **spot openings** (parents start asking for kid-sized tagged bikes), and **build strategies** (raise the tagged price to $100 once demand is confirmed). A business that stops researching after launch is guessing again within months.

Self-check:

- ☐ I listed ≥5 claims my canvas rests on and marked each proven / hinted / belief — EK 2.3.A.1
- ☐ I can say what my **existing** data cannot answer — EK 2.3.A.2.i–ii
- ☐ I gave a verdict on all three gates and know desirability comes first — EK 2.3.A.3.i–iii

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SECTION 2: UNIT 2 TASK 4 (first half) — SECONDARY RESEARCH: THE COMPETITIVE LANDSCAPE

AP Topic 2.3 · LO 2.3.B · EK 2.3.B.1, 2.3.B.2, 2.3.B.3 · CED Unit 2 Task 4

Goal: describe the market you are in **without spending money on new data** — and find the holes that only primary research can fill.

2A. Use ≥4 secondary sources, and cite them like research

#	Source	Type (government / commercial / academic / internal)	Publisher + date	Where (URL / call number)	How current is it?	What you took from it
1						
2						
3						
4						

> **Requirement: at least one must be government or academic** (census / statistics bureau, labor or trade data, industry reports, a university study or published journal). Commercial and social sources are allowed but a 小红书 trend post is not evidence of market size — say what each source is actually good for.
>
> **The CED's five uses of secondary research** (EK 2.3.B.2) — make sure your four sources together cover all five:
> **(1) PESTEL factors** (regulation on reselling, the e-bike trend) · **(2) market size** (how many used bikes sell per month in your city) · **(3) trends** (rising demand for inspected/refurbished goods) · **(4) customer segments** (commuters vs. recreational vs. parents) · **(5) competitor analysis** (what others charge, how they position, what they lack).

2B. The landscape table — this is the "Market" deliverable

List your **real** competitors (direct and indirect, from Assignment 1A), now with data instead of impressions:

Competitor	Price range	How they position	What proof do they offer?	What they lack	Your opening

Model answer from the lesson — Theo's secondary sweep (60 marketplace listings, three fields per listing):

Field	What he found
Price	Range 55–110; **median $82**
Days listed	Cheaper bikes sit longer; **tagged bikes sell faster**
Inspection documented?	**Only 8 of 60** mention any inspection

Without spending a dollar on new data, Theo now knows: almost nobody documents an inspection, and there is a price gap he can fill. **That is a competitive-landscape finding, and it cost him one evening.** Note the shape — 60 units, three fixed fields, countable. Yours must be that countable.

2C. Cost/benefit — be honest about what this section is worth

	Your situation
What secondary got you (fast, cheap, broad)	
What it could not answer (outdated? wrong population? not your question? someone else collected it?)	
Confidence in each figure above (high / medium / low, and why)	

2D. The gap list → this becomes Task 3

Secondary research's real product is **the question you now have to ask in person.**

Gap left by secondary research	The question primary research must answer	Which method (Section 3)

Self-check:

- ☐ ≥4 cited secondary sources, ≥1 government or academic — EK 2.3.B.1
- ☐ My sources collectively cover PESTEL / market size / trends / segments / competitor analysis — EK 2.3.B.2

- ☐ I state what is **stale or unfit** in each number I used — EK 2.3.B.3
- ☐ I left with ≥2 gaps that only new data can fill — that is the bridge to Task 3

SECTION 3: UNIT 2 TASK 3 — HYPOTHESES + METHOD SELECTION

AP Topic 2.3 · LO 2.3.C · EK 2.3.C.1, 2.3.C.2, 2.3.C.3, 2.3.C.4, 2.3.C.5, 2.3.C.6, 2.3.C.7 · CED Unit 2 Task 3

3A. Write two hypotheses — strict form

> **Template:** "If we _____ (change one variable), then ____**%** of _____ (the named population) will _____ (the measured behavior) within _____ (time frame / sample)."

Hypothesis 1 (customer preference — required):

> _____
>
> **Success criterion, set NOW before collecting anything:** _____
> **What we will do if it fails:** _____

Hypothesis 2 (a second preference, or the 2.2 tactic's wording):

> _____
>
> **Success criterion, set NOW:** _____
> **What we will do if it fails:** _____

> **Model — Theo's:** assumption = *buyers will pay more for a documented inspection*; testable form = "*If I add an inspection tag, at least 60% of bikes will sell within one week at $95.*" Notice it has a **variable** (the tag), a **measure** (sell rate), a **number** (60%), and a **clock** (one week). All four, or it is a wish.
>
> **Hypothesis ≠ finding.** The hypothesis is the assumption you feed **in**; the finding is the result handed **back** ("*Version B was preferred 70/30*"). Write the finding only in Section 4 — never in the same sentence as the hypothesis.

3B. Choose the method from the data the hypothesis needs (EK 2.3.C.3–C.7)

Method	Data type	Best for	Weakness	Right for my H1 / H2?
Survey	Quantitative	Preferences, satisfaction, demographics across a **large** group; fixed questions, little follow-up	People may not answer honestly (social desirability bias)	

Method	Data type	Best for	Weakness	Right for my H1 / H2?
Interview (1-on-1)	Qualitative	Deep motivation; open-ended **with follow-up**	Small sample, not statistically generalizable	
Focus group (6–10, moderated)	Qualitative	Attitudes, reactions, generating ideas; adds **group dynamics**	Loudest voice distorts; not a count	
Observation	Behavioral	What people **actually do**, in natural settings, without intervening	Tells you *what*, never *why*	
Experiment	Behavioral	Cause and effect — change one variable, hold the rest constant	Hard to control everything in the real world	
A/B test	Behavioral, real-world	**Will they actually buy?** Two versions, one variable, defined metric	Needs enough units; short windows mislead	

Write the justification, not the choice. For each hypothesis: *the data I need is ____ (quant / qual / behavior), therefore the method is ____, and the method I rejected is ____ because ____.*

Theo's method chain, in order: secondary sweep (60 listings, free) → survey (200 local commuters, "how important is a documented inspection, 1–5") → **A/B test**: 17 bikes, **9 tagged at $95 vs. 8 untagged at $75**, two consecutive Saturdays, same location, same display style → **12 follow-up interviews** with tagged-bike buyers. Quantitative found the pattern; qualitative explained it.

3C. Build the instrument — and defang every question

Write the actual questions / the actual two versions you will show. Then audit them.

#	Your question or version	Leading? (Y/N)	If Y — the neutral rewrite
1			
2			
3			
4			

The biased → neutral model, from the lesson:

Biased (leading)	Neutral
"Don't you think inspected bikes are safer?"	"How important is a documented inspection when buying a used bike?"

Biased (leading)	Neutral
"Wouldn't you prefer a bike with a warranty tag?"	"Which features matter most when choosing a used bike: price, inspection tag, brand, or appearance?"
"Our bikes are the best value — agree?"	"How would you rate the value of this bike relative to its price?"

Rewrite rule: delete "*don't you think,*" "*wouldn't you prefer,*" "*agree?,*" and every superlative. **Offer the options yourself and let them rank.** A leading question doesn't collect an answer, it returns your own.

Theo's interview question that worked: "*What influenced your decision today?*" — not "*Didn't the tag make you feel safer?*" Which would you trust?

CAUTION — if you run an A/B test, name the control explicitly: which **one** variable changed, what stayed identical (location, display, price if irrelevant, time window), and what metric decides the winner. **Two Saturdays is a short window** — the lesson's own verdict on Theo's 40-point gap was "*suggestive, not conclusive.*" Duration and sample are part of the design, not an afterthought.

3D. The sample plan (this is where 2.3.C.8 is decided)

Line	Your answer
The **population** my hypothesis is about (from Assignment 2.1 Section 2 Step 3 — name it)	
Who I will **actually** reach	
Target **n** (number of respondents/units)	
How they are recruited (not "people who happened to reply")	
Sample-bias red flag in my plan (convenience / voluntary response / undercoverage)	
What I will do about it	

The three red flags (EK 2.3.C.8): **convenience sampling** (only your friends), **voluntary response bias** (only the passionate reply — and fill-a-survey-for-a-freebie produces *garbage in, garbage out*), **undercoverage** (a key segment missing entirely — surveying parents about a product kids choose).

The trade to state honestly: "*Quality of the question matters more than quantity of responses.*" 1,000 biased answers < 15 neutral ones from the right population. If your n is small, **say so in one sentence and drop your confidence claim accordingly.** That sentence is worth more marks than the n.

Self-check:

- □ Two hypotheses with variable + measure + number + clock, written **before** data collection — EK 2.3.C.2
- □ Success criteria and the if-it-fails action fixed in advance — EK 2.3.C.2 (anti-post-hoc)

- ☐ Method chosen **from the data type**, with a rejected method justified — EK 2.3.C.3–C.7
- ☐ Zero leading questions left in the instrument after the rewrite pass — EK 2.3.C.8
- ☐ Sample plan names population, actual reach, recruitment, and its own bias risk — EK 2.3.C.8

SECTION 4: RUN IT — FINDINGS, VERDICTS, AND WHAT CHANGED

AP Topic 2.3 · LO 2.3.C · EK 2.3.C.8, 2.3.C.9 · CED Unit 2 Task 3 deliverable

This section is the asterisked deliverable. It cannot be written in advance.

4A. Fieldwork log

Date	Method	Units / respondents	Completed / refused	Notes on conditions (weather, venue, mood, who recruited)

> **Attach your raw data** — the survey export, the interview notes, the tally sheet. Findings without raw data are claims wearing a lab coat.

4B. Results — always with the base

Hypothesis	Measure	Result (with base: __ of __ = __%)	Success criterion	Supported / Not supported / Partially
H1				
H2				

> **Model — Theo's A/B test:**

Group	Sold	Total	Sell rate
Tagged ($95)	7	9	**78%** (7/9 = 77.8%)
Untagged ($75)	3	8	**38%** (3/8 = 37.5%)

> A **40-percentage-point gap** — and the tag, not the price, appears to drive it (the tagged bikes were the *more expensive* ones). **Suggestive, not conclusive:** one test, 17 bikes, two Saturdays.

Always report the denominator. 30 out of 120 is 25%; 30 out of 200 is 15%. A raw count without its base is not a finding — it is a rumor.

4C. The qualitative follow-up — *why* the number moved

For each surprising quantitative result, run ≥3 interviews/observations and list the themes **with frequency**:

Theme (their words, not yours)	Frequency (x of n)	What it explains about 4B

> **Model — Theo's 12 interviews with tagged-bike buyers:**

Theme	Frequency
"The tag makes me trust it's not stolen"	**8 of 12**
"I feel safer knowing someone checked the brakes"	**6 of 12**
"It looks more professional"	**4 of 12**

> The survey never surfaced **stolen-bike fear** — that was the actual motivation. "*Quantitative data identifies the pattern. Qualitative data explains it. You need both.*"
>
> **CAUTION — praise is not data.** "*Praise the dish, then never come back.*" Polite agreement is social desirability bias (EK 2.3.C.4, C.8) — record **what people did** (asked a follow-up, took a card, returned, paid) separately from **what they said**.

4D. The verdict — and this is the graded part

> **Finding 1, stated as a number with its base:** _____
>
> **What we changed in the canvas as a result** (segment / profile / tactic / product / price assumption) **and why:** _____
>
> **The one result we did NOT expect:** _____
>
> **What this test CANNOT claim** (correlation ≠ causation, sample size, short window, seasonality, who recruited them): _____
>
> **The next question this raises:** _____

> **Negative and surprising results are the point.** The CED is explicit that businesses cannot always accurately predict customer or market response and must sometimes innovate with **limited, unclear, or contradictory data** (EK 2.3.C.9). "*We tested it and everyone loved it*" is the weak answer. "*Two of three named price before safety, so we moved the profile from safety-first parent to budget-first parent*" is the strong one — and it is the same sentence pattern that gets the *Exam-Day Validation FRQ* marks.
>
> **Confidence ladder — locate yourself on it** (EK 2.3.C.9):

Level	What it gives you
No research	Pure guess
Secondary only	Landscape understanding

Level	What it gives you
Secondary + survey	Quantitative preferences from your target market
+ interviews	Numbers + motivations
+ A/B test	Numbers + motivations + **real-world behavioral data**

> **Each layer adds confidence. None guarantees the future.** 200,000 data points and $12 million said *launch*. The market said *no*.

Self-check:

- ☐ Every percentage in my report carries its **base** — EK 2.3.C.8
- ☐ I paired a **quantitative pattern** with a **qualitative explanation** — EK 2.3.A.2.i–ii
- ☐ My verdict says **supported / not supported** against a criterion I set in Section 3, not invented now — EK 2.3.C.2
- ☐ I wrote the sentence about what my test **cannot** claim — EK 2.3.C.9
- ☐ At least one canvas element **changed** because of a finding

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SECTION 5: NEW COKE AUDIT — Three Mistakes, Applied to You

AP Topic 2.3 · LO 2.3.A, 2.3.C · EK 2.3.A.2, 2.3.A.3, 2.3.C.8 · [[2.3 New Coke - Case Handout]]

Run your own work past the three failures. This is a 5-minute check that saves a semester.

#	The mistake Coke made	Where it could happen to us	What we did about it
1	**Sample design flaw** (2.3.C.8) — a **sip ≠ a full can.** Tiny controlled samples didn't reflect how people actually drink Coke		
2	**Missing the key variable** — focus groups asked about **taste**, never about **emotional attachment**. "*We are not selling a soft drink. We were selling a little tiny piece of people's lives.*"		
3	**Confirmation bias** (2.3.C.8) — the CEO's mind was already made up ("*it was new Coke or no Coke*"), so contradicting data was ignored		

> **The translation, in one line each:** your instrument may be measuring a **sip** when the decision is a **full can** (a 10-second glance at a listing vs. a parent's actual $95 purchase); your best variable may be the one **you never asked about** (stolen-bike fear was not in Theo's survey — the interviews found it); and if you already believe the answer, your questions will quietly be built to confirm it.
>
> **Mistake #3 is the one only you can fix.** Write it here, honestly:

The answer we secretly hoped for: _____
The result that would have made us uncomfortable — and did we report it anyway?
_____

Bonus case, same lesson from the other company: Crystal Pepsi — Pepsi's own research miss, launched, flopped, and revived 30 years later at 3× the price in a supermarket that read its own sales data. **A failed launch doesn't kill the data. It becomes the data for the next cycle** (EK 2.3.A.4: research after launch). And Pepsi's "collect points, win a fighter jet" promotion broke in court because someone assumed what customers would do instead of measuring it: **check what people actually do.**

SECTION 6: TASK 4 (second half) — APPENDIX: DATA VISUALIZATION

AP Topic 2.3 · LO 2.3.D · EK 2.3.D.1, 2.3.D.2, 2.3.D.3 · CED Unit 2 Task 4 + Appendix

One chart, from **your own research finding** in Section 2 or Section 4. This is a permanent Appendix deliverable for the Business Canvas Project.

6A. Choose the chart from the question — not the other way round

Step	Your answer
The finding I want to show (one sentence, with numbers)	
The question a viewer would ask	compare categories? / total + parts? / change over time? / share of a whole?
Therefore the chart type is	bar / stacked bar / line / pie
Why each of the other three would be worse here	

Selection guide (EK 2.3.D.1–D.3): **Bar** = average sale price by seller type (private $68 · shop $85 · certified $98). **Stacked bar** = monthly sales, January's 20 = 12 tagged + 8 untagged. **Line** = average price of inspected bikes over 6 months — **time on the x-axis. Pie** = Theo's sales, tagged 70% vs. untagged 30% — and **stop at 5 slices.**

The trap: "*Don't default to the prettiest chart. Default to the one that makes your pattern obvious.*" If the question is "*which category is bigger,*" the answer is a **bar chart** — the workhorse of the exam. A time series drawn as bars with 20 categories, or a 9-slice pie, loses the marks for interpretation even when the data is correct.

6B. Draw it — every label required

Paste or attach the chart here. **The five-element checklist is graded:**

- ☐ **Title** that states the **finding**, not just the topic ("*Tagged bikes outsold untagged 78% vs. 38%*" — not "*Sales data*")
- ☐ **Axis labels with units** (¥/$, bikes, %, count) — for a pie: slice labels + percentages
- ☐ **The base / n** on the chart or in the caption (*n = 17 bikes, two Saturdays, same stall*)

- ☐ **Source line** — where the data came from (**your** survey, interview log, A/B tally, or the cited secondary source)
- ☐ **One-sentence takeaway** written beneath it, in plain language someone else could repeat

> **Your chart, drawn:**
>
> **The one-sentence takeaway:** _____

6C. The wrong-chart self-audit

If I had instead drawn this…	The pattern would be…	So my choice was right because…
a pie chart		
a line graph		
a stacked bar		

> **Optional extension — the scatter plot.** Plot two measured variables against each other (Theo: **price vs. days listed** for his 60 listings). His revealed a **dead zone above $85**: almost nothing priced above $85 sold within 14 days — **a price ceiling he did not know existed**, discovered without asking anyone anything. A scatter plot is **not** one of the four permitted Appendix chart types, so use it here as supporting evidence, and pick one of the four for the Appendix. (Note what it did for 2.5: Theo now knows $95 needs the tag to clear the ceiling.)

Self-check:

- ☐ My chart comes from **real findings in this project**, not invented data — EK 2.3.D.1
- ☐ The **type** matches the question, and I can say why the other three are worse — EK 2.3.D.2
- ☐ All five label elements are present, including the base and the source — EK 2.3.D.3
- ☐ Someone who has never met my business can read the chart and repeat the finding correctly

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SECTION 7: EXAM DRILL — The Tells

Classify by **content**, not by label. The last four rows are the ones that separate a 5 from a 3.

#	Item	Classification	One-sentence reason (name the tell)
1	A census table of households by income	primary / secondary	
2	Your own 40-response survey of local parents	primary / secondary	
3	Theo's own **past sales ledger** from last season	primary / secondary	

#	Item	Classification	One-sentence reason (name the tell)
4	"72% of respondents prefer flavor A"	quantitative / qualitative	
5	"I like A because it reminds me of home"		
6	8 people discuss a concept for 45 minutes with a moderator	survey / interview / focus group	
7	Watching which rack shoppers touch without asking them anything		
8	Two landing pages, one headline changed, 48 hours of traffic		
9	"*Most people your kid's age already have one, don't they?*"	biased / neutral + the rewrite	
10	22 of 200 surveyed said they'd buy; 5 people replied to your group-chat message because they were bored	What's the **base**, and which bias is in row 2?	
11	"*Customers rated us 4.6/5*" — only people who loved it replied	voluntary response / convenience / undercoverage	
12	Sales by month for a year	bar / stacked bar / line / pie	
13	Market share: you 12%, shop 48%, driveway 40%		
14	Revenue by product **and** by quarter, in the same bars		
15	"*Version A converted 7 of 9; Version B converted 3 of 8*"	Which version wins, by how many percentage points?	
16	A survey says 70% will buy; only 12% did	What does this prove about research?	
17	Coke tested taste with 200,000 people and still failed	Which of the three New Coke mistakes was this?	
18	A product is loved but your team can't source it	Which gate fails: desirable / feasible / viable?	
19	A product is desired and buildable but loses money at every price		
20	You raise prices and the following month drops — same season, same stall	What can you **not** claim from this?	

> **#3 trap:** internal data you already own is **secondary** — it exists, and it was collected for another purpose. **Primary = new data you go get yourself.** "*Secondary = what's already out there. Primary = what you go get yourself.*"
>
> **#16 / #20 share one answer:** research reduces uncertainty, it does not eliminate it — and a change observed without a control is a **correlation**, not a cause (holiday crowd, weather, one rainy Saturday, or 8 more bikes could bend it).

And the free-response rehearsal (this is the FRQ shape):

> *A business ran a survey, found 72% preferred Feature A, and shipped with A. Sales fell.* ***(a)*** *Identify one flaw in the research design.* ***(b)*** *Identify the data type they failed to collect.* ***(c)*** *Recommend one method that would have caught it.* ***(d)*** *Explain why the research reduced uncertainty but did not predict the outcome.*

Answer it here, in four sentences, before submission:

Self-check:

- ☐ I can sort **primary vs. secondary** even when the data is my own
- ☐ I match **method → data type → question** every time, and reject the mismatched one out loud
- ☐ I report every percentage **with its base** and name the bias in any sample I did not control
- ☐ I pick charts by the **question**, and can say what the wrong chart would have hidden
- ☐ I can state the limits of a test I ran myself — **that** is the sentence the CED is really grading

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Grading Rubric

Section	Excellent (A)	Good (B)	Developing (C)	Needs Work (D/F)
1 — Claim inventory (2.3.A)	5+ claims with sources and honest proven/belief status; existing data sorted by type with what it *cannot* answer; three gates verdicted in order	Claims listed; gates addressed	Beliefs asserted; gates skipped or out of order	Not addressed
2 — Secondary / competitive landscape — Unit 2 Task 4 (2.3.B)	4+ cited sources incl. government or academic, with currency assessed; the four uses (PESTEL/ size/trends/segments/competitors) covered; countable landscape table with price, positioning, proof, gap; explicit gap list	Sources cited and landscape table filled; gaps noted vaguely	Sources uncited or all commercial/ social; table is opinion	No secondary research

Section	Excellent (A)	Good (B)	Developing (C)	Needs Work (D/F)
3 — Hypotheses + method + instrument — Unit 2 Task 3 design (2.3.C)	Two hypotheses in strict variable→measure→number→clock form with **pre-set** criteria and a stated if-fails action; method chosen from data type **and** a rejected method justified; every leading question rewritten; sample plan names its own bias risk	Hypotheses testable, method sensible, questions mostly neutral	Hypotheses are wishes; method picked for convenience; leading questions survive	No hypothesis
4 — Findings + verdict — Unit 2 Task 3 deliverable (2.3.C) **[core]**	Raw data attached; fieldwork log with conditions; every result reported **with its base**; supported/ not supported against the pre-set criterion; qualitative themes with frequency explaining the quantitative pattern; canvas change stated; **the "what this cannot claim" sentence written**	Results reported with bases; verdict stated; some qualitative follow-up	Numbers without bases; verdict invented after the fact; no raw data	Not collected / fabricated
5 — New Coke audit (2.3.A, 2.3.C)	All three mistakes located in the team's own work, with the confirmation-bias row answered honestly about the answer they *wanted*	Mistakes mapped generically to the project	Case retold instead of applied	Not addressed
6 — Data visualization — Unit 2 Task 4 + Appendix (2.3.D)	Chart type **derived from the question** with the alternatives dismissed; built from real project findings; all five label elements present (finding-title, axes+units, base/n, source, one-sentence takeaway); readable by an outsider	Correct chart, labels mostly complete	Pretty-but-wrong chart, or missing base/source; decorative	No visualization
7 — Exam drill	All correct with tell-based reasoning; the 4-part FRQ rehearsal complete	Most correct	Several errors	Mostly incorrect

> **[CORE] Section 4 carries the most weight** — it is the asterisked CED deliverable, and it is what FRQ 1 asks about: how you used **hypothesis testing** to inform decision making in your project.

Important Notes

1. **Same business idea, same team, same customer.** You are not researching a new venture — you are testing the claims you committed to in the Unit 1 Final Canvas and profiled in 2.1. If the findings say the idea is wrong, **that is a success condition, not a failure.** Document the revision.

2. **Secondary first, always** (EK 2.3.B.3). Section 2 before Section 3 — cheap reconnaissance sharpens the questions you then pay to ask. Teams that survey before they read arrive at 100 questions nobody needed and a sample nobody can defend.

3. **Pre-set or it doesn't count.** Success criteria (3A), the neutral rewrite pass (3C), and the sample plan (3D) must exist **before** the first response. A criterion written after seeing the data is a story, not a test — this is precisely how confirmation bias destroyed New Coke at 200,000-person scale.

4. **A small honest n beats a large dishonest one.** Fifteen neutral questions aimed at the population you actually named will out-earn 1,000 leading ones sent to friends. State your bias risk and lower your claims to fit — **the limit sentence is what gets the mark** (EK 2.3.C.8, C.9).

5. **The Appendix chart is permanent.** It travels with your canvas into 2.4, 2.5, and Unit 3–4, and it is one of the three Appendix elements the CED grades (with the pitch and the income statement, both due later). Build it once, correctly labeled, from **real** findings — do not redraw it from a nicer number later.

6. **This is the bridge between 2.2 and 2.4.** Your 2.2 sales tactic is now a **testable preference hypothesis** (Section 3A, H2 is the natural home for it). Your findings here become the **MVP's success bar in 2.4** — and the value proposition you write there should quote a number from Section 4, not a feeling.

7. **Team logistics:** split fieldwork by member, but every member must be able to explain the two hypotheses, the method choice, and the verdict unaided — the exam-day FRQ is written **individually** about the project you all know.

8. **AP Exam connection:** FRQ 1 (Business Canvas Project Exam-Day Validation, **15% of your score, 25 minutes**) draws on LO **2.3.B** and **2.3.C** and asks how hypothesis testing informed your decisions. FRQ 3 (Business Concept Application) draws on **2.3.B, 2.3.D**. Section 4 and Section 6 are literally that question, pre-written.